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Understanding Evolution Strategies for LLM Reasoning: Broader Coverage than GRPO

Population-Based Weight-Space Optimization Preserves Output Diversity That RL Post-Training Collapses

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Post-training LLMs with reinforcement learning — most commonly via GRPO — improves single-sample accuracy (pass@1) but systematically narrows the model’s output distribution, reducing solution diversity across multiple samples. This paper formalises why Evolution Strategies (ES), a gradient-free optimizer that perturbs *weights* rather than policy outputs, avoids this collapse: ES maintains the distributional breadth that RL destroys. On reasoning benchmarks, ES matches GRPO on pass@1 while outperforming it by up to +14.2 pp on pass@32.

AT A GLANCE

Problem: RL post-training (GRPO) improves pass@1 by concentrating output probability mass, but this collapses pass@k diversity — the opposite of what discovery-oriented applications need.

Why it matters: In math, science, and code, generating k diverse candidate solutions and checking the best is often more effective than optimising a single best guess.

Method: Evolution Strategies perturb model weights via isotropic Gaussian noise, estimate a smoothed reward gradient over a population, and update parameters without ever conditioning on per-prompt output distributions — so entropy cannot collapse.

Result: ES matches GRPO within 1–2 pp on pass@1; outperforms by +6.3 pp (MATH-500) to +14.2 pp (AMC/AIME) on pass@32.

Evidence: ES-trained models retain 94% of base model output entropy vs. 71% for GRPO; unique solution fingerprints are $2.3\times$ more frequent at $k=8$.

The Big Picture

Post-training is now the dominant mechanism for improving LLM reasoning. Group Relative Policy Optimisation (GRPO) and its relatives reward high-scoring outputs and penalise low-scoring ones, using group statistics as a variance-reduction baseline. The resulting policy update increases the log-probability of high-advantage outputs — and in doing so, mechanically compresses the output distribution.

For pass@1 tasks — pick the single best answer — this is fine. For pass@k tasks — check at least one of k samples — compressed distributions are harmful: the model generates variations of the same high-reward solution rather than exploring the solution space.

Evolution Strategies sidestep this entirely by never touching output distributions. Instead, ES evaluates perturbed *weight* vectors and updates the base parameters in the direction of higher-reward perturbations. The output distribution at any fixed θ is unchanged by the ES mechanism; only the weight location shifts.

Why Existing Approaches Fall Short

GRPO samples G rollouts per prompt, computes group-relative advantages $\hat{A}_i = (R_i - \bar{R})/\sigma_R$, and applies:

$$\nabla_{\theta} \mathcal{L}_{\text{GRPO}} = - \sum_i \hat{A}_i \nabla_{\theta} \log \pi_{\theta}(a_i|x)$$

This directly increases $\log \pi_{\theta}(a_i|x)$ for high-reward outputs, reducing $H(\pi_{\theta}(\cdot|x))$ each step. Over training, output entropy converges to a low plateau regardless of the base model’s initial diversity. Ablations confirm this: GRPO trained for T steps reduces entropy to 71% of the base model’s, independent of T beyond early training.

PPO has the same problem: the clipped probability-ratio objective $\min(\rho_t \hat{A}_t, \text{clip}(\cdot) \hat{A}_t)$ increases log-probabilities of advantaged actions, compressing distribution mass similarly.

SFT also concentrates distributions, but only on demonstration patterns — it does not optimise diversity directly.

Core Mechanism

ES treats the model as a black box and optimises in weight space. At iteration t , with current parameters θ_t :

1. Sample N noise vectors $\epsilon_i \sim \mathcal{N}(0, I)$
2. Evaluate perturbed models: $R_i = R(\theta_t + \sigma \epsilon_i)$
3. Update: $\theta_{t+1} = \theta_t + \frac{\sigma}{N} \sum_i R_i \epsilon_i$

The update is an unbiased estimate of $\nabla_{\theta} \mathbb{E}_{\epsilon}[R(\theta + \sigma \epsilon)]$, the gradient of the *smoothed* objective.

Crucially, neither the update rule nor the reward evaluation conditions on any per-prompt output distribution $\pi_{\theta}(\cdot|x)$. Output entropy is regulated by the landscape

of weight perturbations, not forced to collapse by logit reinforcement.

Technical Formulation

The ES gradient estimate over population size N is:

$$\hat{g}_{\text{ES}} = \frac{1}{N\sigma} \sum_{i=1}^N R_i \epsilon_i$$

with variance $\text{Var}(\hat{g}) = \frac{\mathbb{E}[R^2]}{N\sigma^2}$, decreasing in N . The paper shows that the smoothed objective $F_\sigma(\theta) = \mathbb{E}_\epsilon[R(\theta + \sigma\epsilon)]$ satisfies:

$$F_\sigma(\theta) \geq F_0(\theta) - \frac{\sigma^2}{2} L_R$$

where L_R is the Lipschitz constant of R , bounding the gap between the smoothed and original objectives.

For a fixed prompt x , the output entropy $H(\pi_{\theta+\sigma\epsilon}(\cdot|x))$ averages over the perturbation distribution rather than being targeted directly. The expected entropy after one update is:

$$\mathbb{E}[H(\pi_{\theta+\hat{g}}(\cdot|x))] \approx H(\pi_\theta(\cdot|x)) + O(\alpha\sigma\|\nabla H\|)$$

compared to the GRPO decrease of $O(\alpha \sum_i \hat{A}_i)$ per step, which is systematically negative for positive-reward outputs.

Learning or Inference Procedure

Training with ES proceeds as:

1. For each iteration, sample a batch of reasoning problems
2. Sample $N = 32$ noise vectors, evaluate all perturbed models in parallel on the same batch
3. Compute ES gradient and update θ
4. Anneal σ by a factor of 0.995 per iteration

No rollout groups, no critics, no probability-ratio clipping. At inference, only θ is used; the noise ensemble is discarded.

What the Guarantee Says

The paper proves that ES converges to a stationary point of the smoothed objective F_σ at rate $O(1/\sqrt{T})$ under standard smoothness assumptions. The output-entropy preservation result is proven for a single gradient step and validated empirically over full training runs.

Experimental Findings

Evaluated with Qwen2.5-7B on MATH-500, AMC/AIME, and LiveCodeBench:

- **pass@1:** ES within 1–2 pp of GRPO across all benchmarks

- **pass@8:** ES outperforms GRPO by +6.3 pp (MATH-500), +11.4 pp (AMC/AIME), +8.7 pp (LiveCodeBench)
- **pass@32:** Advantages grow further; AMC/AIME gap reaches +14.2 pp
- **Output entropy:** ES retains 94% of base model entropy; GRPO retains only 71%
- **Solution diversity:** $2.3\times$ more unique solution fingerprints under ES at $k=8$

Ablations and Interpretation

Population size N : Larger N reduces gradient variance and improves both pass@1 and pass@k, with diminishing returns beyond $N=32$. Smaller N (<16) produces unstable training.

Perturbation scale σ : Larger σ increases exploration (better pass@k) at the cost of slower convergence on pass@1. $\sigma=0.05$ provides a good balance.

Comparison to SFT: SFT with demonstration data improves pass@1 comparably to GRPO but also degrades pass@k, confirming that distribution narrowing is a general property of supervised objectives, not just RL.

Compute cost: ES requires N forward passes per iteration vs. G for GRPO. At $N=G=32$, wall-clock time is comparable because GRPO requires both forward and backward passes while ES requires only forward passes (no gradient through the model).

Reference

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